

Islam Tarek Moawad

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EXPERIENCE

 IEEE-HSB ComSoc <i>Analog IC Design instructor</i>	Oct. 2025 – Present Dokki, Giza
 Marine scout family @ FEHU <i>Member</i>	Oct. 2024 – May. 2025 Cairo, Helwan
Batch leader <i>Batch 61 FEHU leader</i>	Sep 2023 – Feb 2025

EDUCATION

Faculty of Engineering in Helwan, Capital University <i>Junior Electronics and Communications</i>	2023– 2028 Cairo, Helwan
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SKILLS

Languages : Python, C, C++, C#, MATLAB, Bash, Git, $\text{\LaTeX}$
Tools : Atmel Studio, Arduino IDE, ESP32-s3 DevKit, Visual Studio Code, proteus, Multisim, Qucs-s, SimulIDE, KiCAD, LTspice, Cadence Virtuoso, Cisco Packet Tracker, LapView, MATLAB, Logic Friday, Wave Terminal, VMware, Virtual Machine Manager(QEMU/KVM), TeXstudio, WPS Office, Unity Engine, Krita, KDenlive, GitHub, OBS, Super Productivity
Operating Systems I'm familiar with : Windows, Debian, Ubuntu, Kali Linux, Arch Linux

Certifications

Mircococontrollers systems, NTI summer training program	2025 Aug. 2014 – May 2018
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PROJECTS

Google Drive Automation For Academic Materials

- A Python-based automation system that relies on Google services (Google Cloud, Google Sheets API, Google Drive API, Google Forms) to organize summaries and academic materials for my Batch
- It took two weeks to build:
 - * 1st week: Learning Python and Google Cloud
 - * 2nd week: Building the system
- 200 files were uploaded using this system, and the files shared on it remained for newer batches.

ECG R-Pulse Detector

- This is the project I'm most proud of.
- This project was required for the Analog Electronics course.
- No specific ideas were given, so I chose to do the most challenging possible project.
- The project duration was two months and I reached the correct electronic circuit three days before the deadline.

Metro Ticket Finite State Machine

- This was a project in the Logic Circuits course for a team of 10 people.
- I led the team and designed the most complex logarithmic state machine in my batch myself in half an hour.

AM Radio Tx and Rx

- in this project i designed with my team an AM radio system with a PCB implementation of the transmitter and the receiver.

Large Smart Home System With 17 Atmega32

- this was my graduation project of the NTI Microcontrollers systems internship.
- It was a stressful project, more than a difficult one; I learned patience from it.

Analog Implementation Of Neural Networks

- This was one of the first ideas that came to me after learning the basics of Neural Networks.
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- The project was carried out using a set of operational amplifiers, and it was one of the most stressful projects I've ever undertaken.

ROV Control And Communication System

- I am currently working with a team to build an ROV, and I am responsible for the control and communications aspects.

SOFT SKILLS

Languages : Python, C, C++, C#, MATLAB, Bash, Git, $\LaTeX$

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